

A simulation environment for market making under adverse selection, and how learned and closed-form controls behave in it

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Abstract

We extend the `mbt_gym` simulator so that a market maker’s executions carry information about where the price is going next, and we measure how far the Avellaneda–Stoikov and Cartea–Jaimungal controls degrade as that effect is strengthened. Five reinforcement learning algorithms are trained in the same environment on a feature set that does not contain the latent factor. Two results follow. The closed forms lose most or all of their value once the effect is strong, while the learned agents keep roughly two fifths of theirs. More importantly, the learned agents hold terminal inventory almost fixed across an eightfold change in the strength of the effect, whereas the derived closed forms let it grow by a factor of three to ten. Terminal inventory is what the criterion prices, so this is the binding constraint of the problem rather than a secondary statistic.

1 The control problem

A market maker quotes a bid at $S_t - \delta_t^-$ and an ask at $S_t + \delta_t^+$ around a reference price S_t , and is filled when an incoming market order matches one of the quotes. Write q_t for the inventory carried, X_t for cash, and N_t^+, N_t^- for the counting processes of filled sell and buy limit orders. Inventory and cash follow

$$dq_t = dN_t^- - dN_t^+, \quad (1)$$

$$dX_t = (S_t + \delta_t^+) dN_t^+ - (S_t - \delta_t^-) dN_t^-. \quad (2)$$

The objective is the criterion of Cartea et al. [2015]:

$$J = \underbrace{X_T + q_T S_T}_{\text{terminal wealth}} - \varphi \int_0^T q_t^2 dt - \alpha q_T^2, \quad (3)$$

with T the horizon, $\varphi \geq 0$ the price of carrying inventory through time, and $\alpha \geq 0$ the cost of liquidating whatever position is left at T . The last term is a liquidation cost under linear price impact: unwinding q units degrades the average execution price by αq and therefore costs αq^2 .

Two properties of the resulting optimal control matter throughout. Under the assumptions it is derived from, the optimal depths are

$$\delta^{\pm,*}(t, q) = \frac{1}{\kappa} + h_q(t) - h_{q \mp 1}(t), \quad (4)$$

where κ is the decay rate of the fill probability and h solves a linear system indexed by inventory. The control therefore depends on (t, q) and on four constants, and

Table 1: Parameters. The first block is the market, the second the criterion, the third the three mechanisms described in Section 2.1. Only γ is varied.

symbol	meaning	value
T, n	horizon, steps	1.0, 200
S_0	initial price	100
σ	price volatility	2.0
λ	order arrival rate per side	140
κ	fill probability decay	1.5
Q	inventory bound	20
φ	inventory carried	0.01
α	inventory left at T	1.0
γ	signal strength	varied
p	probability of being served	0.95
ζ	own trade price impact	0.1

on nothing else. In particular it carries no state through which an exogenous signal could enter. This is not a defect of the formula; it is its scope, and it is the reason a degradation study is informative.

2 The simulated market

The base simulator is `mbt_gym` [Jerome et al., 2022]. Table 1 lists every parameter used. Prices are continuous, one fill moves inventory by one unit, and the horizon is divided into 200 steps.

2.1 Three mechanisms

Informational adverse selection. This is the object of the study. A mean reverting factor α_t , unrelated to the constant α of Eq. (3), drives both the price drift and the direction of order flow:

$$dS_t = \alpha_t dt + \sigma dW_t, \quad (5)$$

$$\lambda^{\text{sell}}(\alpha_t) = \lambda e^{-\gamma \alpha_t}, \quad \lambda^{\text{buy}}(\alpha_t) = \lambda e^{+\gamma \alpha_t}. \quad (6)$$

When the factor is positive the price drifts up and buy orders become more frequent. Those orders lift the maker’s ask, so it sells just before the rise. The correlation between fills and later price moves is not imposed: it follows from the two channels sharing one factor. Setting $\gamma = 0$ recovers independent Poisson arrivals exactly, which makes the classical environment a limiting case of this one.

Queue position and own trade impact. Two further mechanisms are present but held fixed. With probability $1 - p$ a matching order does not reach the maker’s quote, which stands for not being first in the queue. And each fill moves the reference price by ζ in the direction of the aggressor, so that the maker is pushed against the position it has just taken. Both are included for realism rather than as objects of study; the second is an optimal execution effect and is not what this report is about. Measured against a fixed reference policy, the queue mechanism costs about 5% of profit at $p = 0.95$ and the impact mechanism about 6% at $\zeta = 0.1$. A third mechanism available in the code, in which the price trading through a resting quote forces a fill [Lalor and Swishchuk, 2024], is left switched off: at the depths quoted here such an event is close to a five sigma move and the mechanism is numerically inert.

3 Agents

3.1 Closed forms, and their refitted counterparts

Two analytic controls are used. Avellaneda–Stoikov [Avellaneda and Stoikov, 2008] maximises the exponential utility of terminal wealth and reads a risk aversion, a volatility and a fill decay. Cartea–Jaimungal [Cartea et al., 2015] maximises Eq. (3) directly and reads κ , λ , φ and α . Deployed unchanged in the enriched market, each is simply misspecified, and what it loses measures the cost of that misspecification rather than the value of anything that might replace it.

Each therefore also appears in a refitted form: the same functional shape, with its own internal constants chosen by grid search on simulated returns, refitted separately at every signal strength. No parameter is added, and the objective stays pinned to the environment’s reward so every arm is scored identically. This arm should be read as an upper reference rather than as a competitor. It runs its search inside the true environment, so it knows both the functional form and the answer, which a deployed strategy would not. The search is scored on three fitting seeds and reported on four disjoint evaluation seeds, because an argmax over a noisy objective buys part of its own improvement.

3.2 Learned agents

Five algorithms are used, chosen to span the families the two recent surveys organise around [Bai et al., 2024, Ghasemi et al., 2025] rather than to collect recent methods. Table 2 lists them. Reinforcement learning has been applied to market making since Chan and Shelton [2001], whose state variables are inventory, order imbalance, spread and recent price changes; the feature set below was arrived at independently and is very nearly the

Table 2: The five algorithms and why each is present. All share a two layer network of width 64 with ReLU activations and the discount fixed to one, since the criterion is an undiscounted finite horizon sum. Everything else is left at library defaults.

algorithm	role
PPO [Schulman et al., 2017]	on policy with a trust region; the usual reference in this literature
A2C [Mnih et al., 2016]	the same without the trust region, which isolates what clipping buys
SAC [Haarnoja et al., 2018]	off policy, maximum entropy; sample efficient
TD3 [Fujimoto et al., 2018]	off policy, deterministic policy, twin critics
Recurrent PPO	PPO with an LSTM; the factor is latent, so the sufficient statistic is a function of history

same.

What the agent sees. The latent factor is never given to a learned agent. Its features are inventory, time remaining, the imbalance and the total count of recent order arrivals, the imbalance of its own recent fills, and the return over a recent window. All are quantities a market maker observes. Two windows are used, five steps for the count based features and twenty for the return, because the first is limited by arrival noise and the second by how fast the factor itself moves; the two optima were measured, not assumed.

4 Method

Nesting. Every mechanism has a setting at which the enriched environment must reproduce the classical one trajectory by trajectory, not merely in distribution. This is asserted by test, and it is what makes the analytic control a limiting case rather than an out of distribution baseline whose gap would be uninterpretable.

Replication. A learned arm is trained on three seeds and each trained policy is evaluated on four further seeds, disjoint from the training ones and shared across all arms, with 1500 trajectories per seed. The spread reported for a learned arm is taken across training seeds, since two runs of one algorithm differ and that is the variability that matters when comparing algorithms. For an analytic arm the spread is taken across evaluation seeds. Every arm sees the same evaluation seeds, so comparisons between arms are paired.

How degradation is measured. An arm is first measured in the environment with the mechanism switched

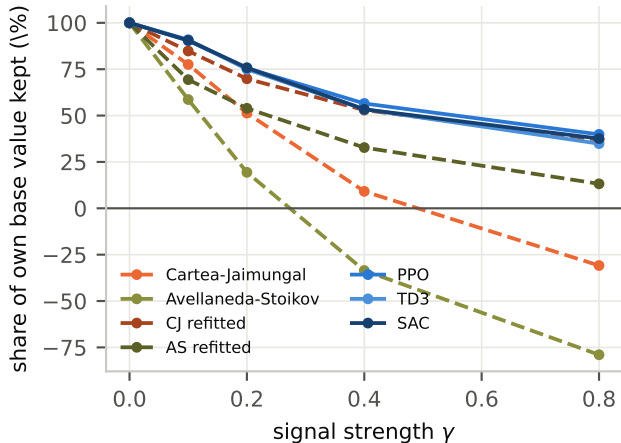


Figure 1: Share of its own reference value that each arm retains. Solid lines are learned agents, dashed lines are analytic controls. The horizontal line at zero separates arms that still earn from arms that lose. Reading the right hand edge: the learned agents and the refitted Cartea–Jaimungal keep between a third and two fifths, while the two derived controls are below zero.

off, which gives it a reference value. The mechanism is then strengthened and the arm is measured again, unchanged. We report the share of its own reference value that it retains. An arm that scored 60 and now scores 24 has kept 40% of what it had. This is a within arm comparison, so it does not reward an arm for having been strong to begin with.

Quantities reported. Alongside the criterion of Eq. (3) we report four quantities that do not depend on how the criterion prices risk: terminal wealth, its standard deviation across trajectories, the mean absolute inventory held at the horizon, and the number of orders filled per episode. A change visible only in the criterion is a statement about the scoring; a change visible in these four is a statement about behaviour.

5 Results

5.1 Value kept as the effect strengthens

Figure 1 shows, for each arm, the share of its own reference value retained as the signal strength rises from zero to 0.8. The two analytic controls fall below zero: at the strongest setting Cartea–Jaimungal ends at -31% of its reference value and Avellaneda–Stoikov at -79% , meaning both destroy value rather than merely earning less. The refitted controls and the three learned agents stay in a narrow band between 13% and 40%.

Table 3 gives the full picture at the strongest setting. The criterion column separates the arms into those that still work and those that do not. The four columns to its

Table 3: All arms at signal strength 0.8. Criterion is Eq. (3); wealth is its first term; the last column is orders filled per episode, with the share of the same arm’s own count at zero signal in parentheses.

arm	crit.	wealth	vol.	$ q_T $	orders
Cartea–J.	−18.4	25.1	29.2	4.39	74 (76%)
Avell.–S.	−38.9	26.6	13.5	6.74	60 (65%)
CJ refit.	22.1	30.1	13.2	1.79	41 (43%)
AS refit.	6.5	22.5	7.0	3.35	31 (34%)
PPO	22.1	23.4	7.6	0.81	25 (32%)
TD3	19.5	20.3	6.8	0.59	31 (38%)
SAC	21.0	22.4	7.7	0.85	35 (39%)

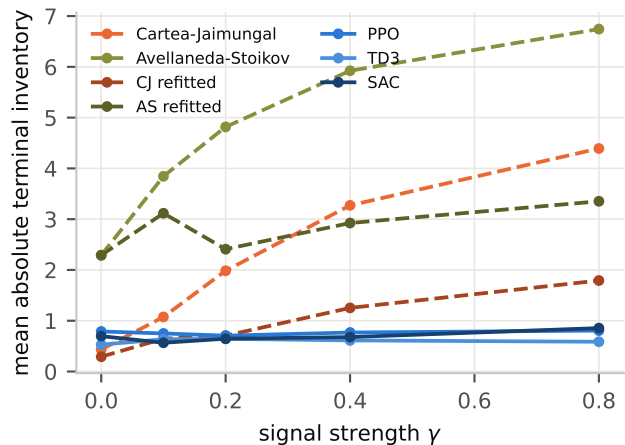


Figure 2: Mean absolute inventory held at the horizon. The three learned agents are the flat lines near the bottom. Over an eightfold increase in signal strength their terminal inventory changes by at most a quarter, while the derived Cartea–Jaimungal control multiplies its own by ten.

right are independent of how the criterion prices risk.

5.2 Terminal inventory

The criterion prices whatever position is left at the horizon, so holding that position near zero is the binding constraint of the problem rather than a secondary statistic. Figure 2 tracks it.

The learned agents hold terminal inventory between 0.53 and 0.86 units at every signal strength. Over the same range the derived Cartea–Jaimungal control goes from 0.43 to 4.39, a factor of ten, and the refitted one from 0.29 to 1.79, a factor of six. Avellaneda–Stoikov never reads the reward function at all, so its policy is identical at every setting and its inventory rises to 6.74 purely because the market changed underneath it.

This is the clearest separation in the study, and it is not close. At the strongest setting the learned agents end with two to eight times less inventory than any analytic arm, and they do so while giving up less wealth per unit

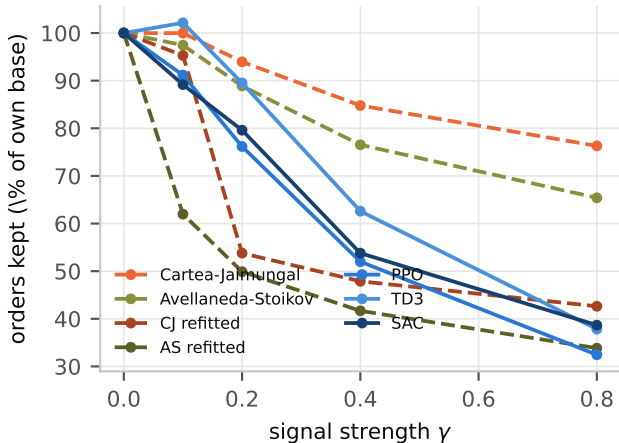


Figure 3: Orders filled per episode, as a share of the same arm’s count at zero signal. Every arm that still earns value cuts activity to between a third and two fifths. The arm that keeps trading most is also the one that ends worst.

of its standard deviation.

5.3 Activity

A control can always reduce what adverse selection costs it by quoting wider and trading less. Whether the arms do that is visible in the last column of Table 3 and in Figure 3.

All arms that retain value converge to a similar level of activity, between 32% and 43% of their own base count. The learned agents are inside that band, not above it: they are not more active than the refitted control. What the figure does show is that activity is not by itself a virtue. The derived Cartea–Jaimungal control keeps 76% of its orders, the highest of any arm, and it is the arm that loses the most value. It keeps quoting into flow that has turned against it, which is exactly the failure the mechanism was built to produce.

The right reading is therefore that reducing activity is necessary but not sufficient, and that the arms are separated by what they do with the orders they keep. Per order filled, markout at a five step horizon is -0.10 to -0.12 for the learned agents against -0.17 for the refitted control and -0.21 for the derived one, so the flow the learned agents retain is roughly half as toxic.

5.4 An asymmetry in the protocol

The only arm that matches the learned agents on the criterion is the refitted Cartea–Jaimungal control, and the comparison is not symmetric. That arm searches over its own constants inside the true environment, once per setting, so it is given both the functional form and the answer. The learned agents received no tuning: library default hyperparameters, one network size, no architec-

ture or learning rate search, with only the discount fixed because the criterion requires it. The protocol is tilted toward the analytic side, and the learned agents still match it on the criterion while ending with less inventory and less toxic flow.

6 Limits

The measurements above hold inside one simulated market, and four limits bound what they say.

The quoted spread is wide relative to the volatility. The controls quote about 67 basis points around a price whose one step move is 14 basis points. A liquid futures contract has a spread two orders of magnitude tighter relative to its volatility. The consequence is that the price essentially never trades through a resting quote, so the mechanical channel of Lalor and Swishchuk [2024] is inert and only the informational channel is exercised. Reaching the other regime is a joint constraint on the fill decay, the volatility and the step size, not a setting to switch.

The signal is generous. At one standard deviation of the factor the two arrival intensities differ by a factor of six. Order flow predictability in traded data is far weaker. The strongest setting used here should be read as an extreme case rather than a central one.

Volatility is constant. There are no calm and turbulent regimes, so the ability to widen when the market turns is not tested. A stochastic variance process is implemented and verified to reproduce the constant case exactly when its volatility of volatility is zero, but activating it was measured to change the reference policy’s outcome by under two percent, so it is left off. Making regimes matter would require coupling the variance to the arrival intensity, which is a modelling choice rather than a parameter.

Budget. The learned agents are compared at a fixed number of environment steps. Their criterion is still rising with training at that point, so any ordering read at one budget is a statement about that budget. Longer runs move the learned agents up relative to the refitted control; the point at which they cross it is a property of this configuration and of the compute spent, not a quantity worth reporting on its own.

7 Conclusion

Strengthening the informational component of adverse selection degrades every control in the study, which is expected: neither analytic control carries a state through

which the factor could enter, and this is visible in their formulas. What the measurements add is how differently the arms degrade.

The derived controls do not merely earn less, they stop working, ending below zero once the effect is strong. The learned agents keep about two fifths of their own reference value, which is close to what a control refitted inside the true environment keeps.

The separation that is not close is inventory. The learned agents hold terminal inventory almost fixed across an eightfold change in the strength of the effect, while the analytic controls let it grow by a factor of three to ten. Since the criterion prices exactly that quantity, and since the agents reach it without being given the latent factor and without any hyperparameter tuning, this is the result we would carry forward.

Two things this does not establish. It does not show that a learned agent would hold up in a market calibrated to traded data, since the environment here is not; and it does not separate what the agents learn about the factor from what they learn about inventory control, since both improve together. The natural next step is a policy with oracle access to the factor, which would bound how much of the remaining gap is attributable to prediction.

Reproducibility

The environment, the agents, the training scripts and the evaluation notebooks are in the project repository. Trained policies are regenerated by the training scripts; every number in this report comes from an executed notebook.

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